

ARDHI UNIVERSITY



**RESEARCH TITLE: IDENTIFICATION OF NEW SUITABLE LOCATIONS FOR
REMOTE SUBSCRIBER UNIT PLACEMENT IN DAR ES SALAAM**

A Case Study of Dar Es Salaam

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IDENTIFICATION OF NEW SUITABLE LOCATIONS FOR REMOTE SUBSCRIBER UNIT PLACEMENT IN DAR ES SALAAM

A Case Study of Dar Es Salaam

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A Dissertation Submitted to the Department of Geospatial Sciences and Technology in
Partially fulfilment of the Requirements for the Award of Bachelor of Science in
Geographical Information Systems and Remote Sensing (BSc. GIS & RS) of Ardhi
University.

CERTIFICATION

The undersigned certify that they have proof read and hereby recommend for acceptance of the research entitled “**Identification Of New Suitable Locations For Remote Subscriber Unit Placement In Dar Es Salaam**” and now recommend it for acceptance by Ardhi University.

.....

Ms. Beatrice Kaijage

(Supervisor)

Date.....

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I KELVIN FAUSTIN LIHAWALA declare that this is my own original work and that to the best of my knowledge, it has not been presented to any other University for a similar or any other degree award except where due acknowledgements have been made in the text.

.....

Kelvin Faustin Lihawala

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DEDICATION

This work is dedicated to the people who have stood beside me with unwavering support, love, and belief in my journey.

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ABSTRACT

In the rapidly evolving digital age, access to reliable telecommunication services is essential for socio-economic development. Urban centers such as Dar es Salaam continue to experience increasing demand for improved network coverage and quality. One key solution in meeting this demand is the strategic deployment of Remote Subscriber Units (RSUs) telecommunication nodes that extend connectivity to end users where traditional infrastructure is limited or insufficient. However, identifying optimal locations for RSU deployment poses challenges due to diverse environmental, infrastructural, and accessibility constraints.

This study aimed to analyze and identify suitable sites for the establishment of new RSU stations in Dar es Salaam using Geographic Information Systems (GIS) and Multi-Criteria Decision Analysis (MCDA). A set of spatial datasets comprising Digital Elevation Model (DEM), land use/land cover, road networks, rivers, and building footprints was used to develop a suitability model and a restriction model. The suitability model was created using a weighted overlay method, where slope (60%), land use (25%), and road proximity (15%) were reclassified and integrated to reflect RSU placement preferences. The restriction model excluded areas that are flood-prone, inaccessible, or far from buildings by applying buffers and raster conversion techniques.

The final suitability output was produced by multiplying the suitability and restriction models, resulting in a spatial map that highlights areas both suitable and unrestricted for RSU deployment. Key zones such as Sinza, Mikocheni, Ubungu, and Tegeta were identified as highly suitable. To assess the model's accuracy, eight existing RSU coordinates were used for validation, with results showing that 87.5% of the RSUs fall within moderate to highly suitable areas, confirming the reliability of the approach.

This research demonstrates that GIS-MCDA provides a practical, data-driven framework for RSU site selection. The method supports better decision-making for network expansion, reduces reliance on field surveys, and aligns with infrastructure planning needs in rapidly growing cities. The study recommends the integration of this model into telecommunication planning workflows and encourages future enhancements through expert-based weighting, population analysis, and signal simulation.

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ABBREVIATIONS

RSU	Remote Subscriber Unit
TTCL	Tanzania Telecommunications Corporation Limited
GIS	Geographic Information Systems
DEM	Digital Elevation Model
MCDA	Multi Criteria Decision Analysis
MCDM	Multiple Criteria Decision Making
WSM	Weighted Sum Model
ICTs	Information and Communication Technologies
RS	Remote Sensing
TP&TC	Tanzania Posts and Telecommunications Company
TPC	Tanzania Posts Corporation
TPB	Tanzania Postal Bank
TCC	Tanzania Communications Commission
TCRA	Tanzania Communications Regulatory Authority
FTTC	Fiber-To-The-Cabinet
FTTN	Fiber-To-The-Node
SMCA	Spatial Multi-Criteria Analysis
OSM	Open Street Map
MCE	Multi Criteria Evaluation
WGS 84	World Geodetic System 1984
UTM	Universal Transverse Mercator

CHAPTER ONE

1 INTRODUCTION

1.1 Background

In today's world, communication is at the heart of how people live, work, and connect. From phone calls and internet browsing to mobile banking and online education, reliable communication networks are essential to modern life. Behind these everyday services is a complex system known as a telecommunication network (Alenoghena et al., 2023). Telecommunication network is a network of technologies and infrastructure that allows information to be shared across distances in real time.

To function efficiently, telecommunication networks depend on various components such as cables, antennas, routers, and switching stations. However, in many developing countries, especially in rapidly growing urban areas, these networks often face challenges related to poor infrastructure, coverage gaps, and increased demand (Meena & Geng, 2022). One solution that helps address these challenges is the use of Remote Subscriber Units (RSUs).

Remote Subscriber Unit is a telecommunications device that establishes connectivity between the central network infrastructure and end-user devices, providing seamless communication over long distances (Husby & Vinge, 1985a). RSUs are particularly critical in addressing challenges related to poor network infrastructure or areas experiencing high service demand (Fiorani et al., 2015a). Their deployment plays a vital role in expanding network coverage and ensuring reliable service delivery, especially in remote or underserved areas. (Thomas et al., 2016).

Globally, RSU's have revolutionized the telecommunications industry by enabling efficient network expansion in regions where traditional infrastructure is costly or impractical to deploy (Rajpal & Goyal, 2017). This study aligns with Sustainable Development Goal (SDG) 9, particularly target 9.1, which promotes the development of sustainable and resilient infrastructure, and target 9.2, which advocates for industrial growth and increased employment opportunities (Saini et al., 2023). Furthermore, the deployment of RSUs supports national ICT policies aimed at reducing the digital divide and fostering economic growth through improved connectivity. For example, in rural Canada, telecommunication companies have successfully deployed RSUs to extend network coverage across vast and sparsely populated territories, where laying fiber-optic cables would be economically unfeasible. These RSUs, often integrated with wireless microwave or satellite backhaul, ensure that residents in remote

northern communities have access to voice, internet, and emergency services greatly enhancing connectivity and quality of life.

Locally, RSUs serve as a strategic solution for TTCL to address coverage gaps in underserved regions, enhancing service quality and accessibility for customers (Thomas et al., 2016). By optimizing RSU placement, TTCL can contribute to reducing inequalities in access to Information and Communication Technologies (ICTs), driving sustainable development and innovation in Tanzania. In this context, RSUs offer a cost-effective and scalable method for expanding service delivery. By bridging the digital divide, RSUs play a pivotal role in enhancing socio-economic development through improved access to communication technologies. TTCL, as the national service provider, faces difficulties in delivering consistent and reliable network coverage, especially in remote and rural communities (Beletskiy, 2023)

Existing studies highlight the potential of RSUs in overcoming infrastructural challenges in the telecommunications sector. Researchers emphasize the importance of data-driven methods, such as GIS, in optimizing the placement of RSUs for maximum efficiency and cost-effectiveness (Beaudet et al., 1994) for example a study by(Beaudet et al., 1994) says the service providers are required to place cellular towers at the optimal location for providing a strong cellular network in a particular region. Similarly,(Singh & Jindal, 2017) addresses the problem of optimal distribution of base tower placement while maintaining acceptable quality of service. Despite these findings, most research has focused on global case studies, with limited focus on the Tanzanian context, leaving significant gaps in local applications.

Traditional methods still dominate the decision-making process in Tanzania, leaving room for the adoption of innovative approaches to improve network planning. While there is extensive literature on RSU deployment, limited research exists on the integration of GIS and Remote Sensing (RS) techniques for suitability analysis in the Tanzanian. GIS and Remote Sensing (RS) have been widely used in other various fields, their application in optimizing RSU placement remains underexplored in Tanzania. Study by (Ustaoglu & Aydinoglu, 2020a)demonstrate the potential of GIS in suitability analysis for infrastructure planning, but these methodologies are rarely applied to the telecommunications sector locally. Most studies focus on manual or less sophisticated techniques for identifying suitable locations, neglecting the potential of GIS-based methodologies to enhance precision and efficiency in site selection. This research seeks to address this gap by proposing a GIS-based framework to meet TTCL's needs.

1.2 Statement of the Problem

The rapid expansion of telecommunication networks has increased the demand for reliable and efficient coverage in many regions especially Dar es salaam. As one of the fastest-growing cities in East Africa, Dar es Salaam faces challenges in selecting suitable sites for RSU placement due to its dense population and rapid urbanization. Traditional site selection often relies on simplistic methods of identifying the criteria such as simple observation but also involves a lot of field work (Xu et al., 2023). This research addresses the gap by incorporating a more efficient methodology that combines traditional approaches with GIS to optimize RSU placement in Dar es Salaam. This approach reduces the reliance on extensive fieldwork while ensuring reliable network coverage.

1.3 Objectives

1.3.1 Main objective

The main objective is to identify new locations for Remote Subscriber Unit (RSU) deployment in Dar es Salaam using GIS-based analysis.

1.3.2 Specific objectives

- i. To map key geographic and environmental factors affecting Remote Subscriber Unit station placement e.g. Land use and slope.
- ii. To map the proximity of proposed Remote Subscriber Unit station locations to roads.
- iii. To map optimal locations for new Remote Subscriber Unit station placement.

1.4 Research questions

- i. What are the most suitable places to establish new RSU?
- ii. How far should the proposed RSU be from roads?
- iii. What geographic and environmental factors that affect RSU placement?

1.5 Scope and limitation

1.5.1 Scope

This research explores the importance of remote subscriber units and the areas which are suitable to establish them thus provide internet service to those areas which may help in various activities such as education. The study will base on Dar es salaam as a case study specifically TTCL remote subscriber units though this research could also help other telecommunication service providers to find new suitable areas to establish their infrastructures.

1.5.2 Limitation

- ✓ **Limited Availability of High-Resolution Data**

The elevation datum, was obtained from a global source with moderate resolution (i.e 30 meters). This may have reduced the precision of the final suitability model

- ✓ **Lack of Field Verification**

Due to time and resource constraints, most of the validation was done using existing RSU coordinates rather than physical site visits. This limited the ability to confirm on-the-ground conditions such as land ownership, accessibility, or environmental hazards.

- ✓ **Simplified Assumptions in Weighting Criteria**

The weights used in the multi-criteria analysis were based on logical reasoning and literature but were not calibrated with expert judgment or real cost-benefit data from telecom engineers or TTCL planners. This introduces subjectivity into the suitability scoring.

- ✓ **Technical Constraints of GIS Tools**

While GIS is powerful for spatial analysis, it does not support real-time network performance monitoring or signal simulation. Therefore, the model does not reflect actual signal strength or coverage, only geographic suitability.

1.6 Significance of the research

- i. **Field work reduction**

By identifying suitable areas to establish new RSU stations. This would reduce time taken to complete the task since the field work required is only validating that the proposed area fits the criteria.

- ii. **Cost-Effective Solutions**

The Research may propose strategies for reducing the operational costs of cellular networks, making them more accessible in developing regions.

- iii. **Network Planning Improvement**

By applying a GIS-based Multi-Criteria Decision Analysis (MCDA), this study introduces a data-driven and spatially informed framework that enables more precise, transparent, and scalable RSU placement. The model considers essential criteria such as terrain slope, proximity to roads, land use patterns, and building density all of which influence the effectiveness and accessibility of RSUs.

1.7 Beneficiaries

- i. **TTCL staff:** The research helps TTCL identify suitable locations for new RSU stations, reducing fieldwork, costs, and time spent on site selection.
- ii. **Other Telecommunication Service Providers:** The findings can assist other telecom companies in identifying optimal areas for infrastructure expansion.
- iii. **Customers** – Improved RSU placement will enhance network coverage and service quality, benefiting end-users in Dar es Salaam.
- iv. **Researchers and Academics** – The study adds to the knowledge base on cellular networks and GIS-based infrastructure planning.
- v. **Policy Makers and Urban Planners** – The research aligns with sustainable development goals and ICT policies, contributing to better network planning strategies.

CHAPTER TWO

2 LITERATURE REVIEW

2.1 Overview

This chapter reviews existing literature on telecommunication infrastructure planning, remote sensing and GIS applications in site suitability analysis, it evaluates previous studies on RSU and base station placement, highlighting methodologies, criteria used, and data sources. The chapter establishes the research gap that this study addresses

2.2 Remote subscriber unit

is a remote device that connects end users (subscribers) to the main telecom network(Husby & Vinge, 1985b)Functions of an RSU include acts as a relay between subscriber equipment (like telephones or modems) and the central telecom exchange, handles voice and data transmission to and from users, provides local switching capabilities in some cases, reducing the need to route all calls through the main central office, improves service in remote or under-served areas, such as rural zones or densely built urban environments(Fiorani et al., 2015b) Remote Subscriber Unit (RSU) plays a key role in delivering high-speed communication services over long distances while keeping the infrastructure cost-effective(Ganguli & Heinonen, 1990)

2.3 Tanzania Telecommunications Corporation Limited

Tanzania Telecommunications Corporation, formerly known as Tanzania Telecommunications Company Limited (TTCL), was established through the Tanzania Telecommunications Company Incorporation Act of 1993 following the liberalization of the communication sector.(Sree Laxmi Pavani et al., 2024) The company began operations on January 1, 1994, after the split of the former Tanzania Posts and Telecommunications Company (TP&TC) into four entities: Tanzania Posts Corporation (TPC), TTCL, Tanzania Postal Bank (TPB), and Tanzania Communications Commission (TCC), now known as the Tanzania Communications Regulatory Authority (TCRA).

In line with economic reforms, TTCL was partially privatized in 2001, with a consortium from the Netherlands and Germany acquiring 35% shares. However, the Government of Tanzania regained full ownership of the company on June 23, 2016(Tao et al., 2019) . A new law, the Tanzania Telecommunications Corporation Act No. 12 of 2017, was enacted to restructure TTCL into a public corporation. The Act came into effect on February 1, 2018, marking the

formal establishment of the Tanzania Telecommunications Corporation (TTC), which assumed the responsibilities of the former TTCL.

2.4 Telecommunication network

A telecommunications network is fundamentally composed of transmitters, receivers, and communication channels, all working together to enable the transmission of data. These networks can be broadly categorized into analog and digital systems, each with distinct components such as switches, routers, and repeaters (Altekar et al., 2021). In digital networks, routers are used to intelligently forward information to its destination, while analog networks rely on switches to establish temporary, dedicated paths between users. Regardless of type, repeaters play a crucial role in signal regeneration, especially over long distances, where signal attenuation could otherwise degrade quality.

This foundational structure is directly relevant to the deployment of Remote Subscriber Units (RSUs) in modern telecommunication systems. An RSU serves as a remote access point that connects end-users to the broader network infrastructure (Pospíšilová et al., 2015). It often functions as a mini-switching or signal distribution node, strategically placed at the network edge to bring services closer to subscribers, especially in areas not directly connected to the central office.

In fiber-optic or hybrid fiber-coaxial systems, RSUs help convert and extend digital signals to the subscriber level often incorporating micro-routers to facilitate this function (Fenta et al., 2021). Furthermore, RSUs can house repeaters or amplifiers to maintain signal strength across extended distances, reducing latency and packet loss, particularly in underserved or remote urban areas like some parts of Dar Es Salaam.

2.5 Fiber in telecommunication industry

Fiber optic cables are a fundamental component of modern telecommunications infrastructure, transmitting data as light signals through thin strands of glass or plastic known as optical fibers (Liu et al., 2015). These fibers are grouped into bundles and protected by a durable outer jacket, allowing for long-distance, high-speed, and high-capacity data transmission. Compared to traditional copper cables, fiber optics provide significantly higher bandwidth, lower signal loss, and greater resistance to electromagnetic interference, making them ideal for the backbone of national communication systems (Z. Wang & Xian, 2023).

In the context of Remote Subscriber Unit (RSU) deployment, fiber optic cables play a critical role in enabling efficient signal delivery from central exchanges to remote areas. RSUs are often integrated into fiber-to-the-cabinet (FTTC) or fiber-to-the-node (FTTN) architectures, where fiber optics are used to carry signals close to the end-user, and the RSU acts as an intermediate distribution point (Nyarko-Boateng et al., 2020). From there, the signal is converted and transmitted to subscribers via copper lines, wireless links, or other last-mile technologies. This hybrid setup leverages the speed and reliability of fiber while minimizing deployment costs in densely populated or geographically challenging areas (Meier, 2012).

In urban environments like Dar es Salaam, RSUs connected through fiber backhaul can significantly enhance network performance and coverage, particularly in underserved neighborhoods (Bastianini et al., 2019). Therefore, the integration of fiber optic infrastructure with RSU deployment is essential for expanding access, improving service quality, and supporting the growth of digital communication services across the region.

2.6 Multicriteria Decision Analysis

Multi-Criteria Decision Analysis (MCDA), also referred to as Multiple Criteria Decision Making (MCDM), is a structured approach used to evaluate and prioritize multiple alternatives based on a set of defined criteria. It is especially useful in complex decision-making scenarios where trade-offs must be made between conflicting factors (Thokala et al., 2016). In the context of Remote Subscriber Unit (RSU) site selection, MCDA provides a robust framework for analyzing potential locations by assessing various criteria such as population density, proximity to existing infrastructure, terrain, network demand, cost-effectiveness, and ease of access.

By applying MCDA techniques, telecommunication planners can quantify and compare the suitability of different locations for RSU deployment, leading to more transparent, data-driven decisions. This is particularly relevant in urban environments like Dar es Salaam, where multiple variables technical, economic, and geographic must be carefully balanced (Marsh et al., 2016). MCDA models such as the Analytical Hierarchy Process (AHP) or Weighted Sum Model (WSM) can be integrated with GIS tools to visualize and rank locations spatially, enhancing the efficiency and accuracy of RSU planning.

2.7 Analytic Hierarchy Process (AHP)

The Analytic Hierarchy Process (AHP) is a well-established decision-making technique that supports suitability analysis by helping determine the relative importance of multiple, often competing factors in complex spatial or planning decisions. AHP structures the decision

problem into a hierarchical model, where criteria and sub-criteria are compared in pairs to generate weighted values that reflect their influence on the overall goal (Tripathi et al., 2022). In the context of Remote Subscriber Unit (RSU) site selection, AHP is particularly valuable for evaluating diverse spatial and technical criteria, such as population density, proximity to fiber infrastructure, land use patterns, accessibility, terrain elevation, and service demand.

By integrating AHP with Geographic Information Systems (GIS), planners can conduct spatial multi-criteria analysis (SMCA) to generate a suitability map, highlighting areas that best meet the weighted criteria for RSU deployment. This approach enables objective, data-driven site selection, ensuring that decisions are both technically sound and economically feasible (Darko et al., 2019). In urban environments like Dar es Salaam, where infrastructure planning must consider rapid urbanization, service gaps, and physical constraints, AHP enhances the precision and defensibility of RSU placement strategies. The method's ability to combine expert judgment with empirical data makes it a powerful tool for optimizing telecommunications infrastructure development.

2.8 Euclidean distance

Euclidean Distance is a spatial analysis method used in GIS to calculate the shortest path between two points. It is instrumental in proximity analysis, especially in infrastructure planning (L. Wang et al., 2005). In this study, Euclidean Distance was used to determine the closeness of potential RSU sites to roads and buildings key factors in accessibility and service delivery. Proximity-based criteria help align infrastructure development with real-world operational needs. This distance is reclassified into suitability classes and incorporated into the overall MCDA process (Mesquita et al., 2017).

2.9 GIS in Telecommunication Planning

Geographic Information Systems (GIS) play a vital role in modern telecommunication infrastructure planning. By integrating various spatial datasets such as elevation, population density, land use, and road networks GIS enables planners to make data-driven decisions for optimal infrastructure placement. According to (Castells et al., 1994) GIS-based analysis can enhance decision-making by visualizing geographic constraints and opportunities. (Eiselt et al., 2023) further highlight how GIS supports the efficient placement of towers and network nodes, reducing both cost and signal loss. Despite global advancements, GIS-based RSU planning remains underutilized in Tanzania, justifying the need for this research.

2.10 Validation of GIS Suitability Models

Validation is a critical component in assessing the effectiveness of GIS-based suitability models. One common approach is the use of point-based validation, where the predicted suitability is tested against existing infrastructure locations. In this study, eight existing RSU sites in Dar es Salaam were used for validation. Suitability scores were extracted from the final suitability raster at each RSU location using the "Extract Values to Points" tool. The results showed that most RSUs fell within moderately or highly suitable areas, supporting the model's accuracy. This validation method, supported by (Ferretti, 2011), enhances the credibility of GIS-based planning tools.

CHAPTER 3

3 METHODOLOGY

3.1 Overview

This chapter presents the methods and techniques used to conduct the study. It describes the study area (Dar es Salaam), data sources (e.g. land cover, DEM for elevation, OSM for roads and settlements), and software tools (QGIS, Google Earth Engine). It explains the multi-criteria evaluation (MCE) approach, including criteria selection, weighting, reclassification, and overlay analysis. The data preprocessing steps and suitability model logic are detailed clearly

3.2 description of the study area

The study area is the northern part of Dar es Salaam, which is the area of influence of the a TTCL branch located at Kijitonyama, the branch is known as TTCL North. It was selected as the case study due to its high population density, rapid urbanization, and status as Tanzania's economic hub (country's center of economic activities). These factors contribute to a growing demand for reliable telecommunication services, making it a strategic location for analyzing the suitability of new remote subscriber units.

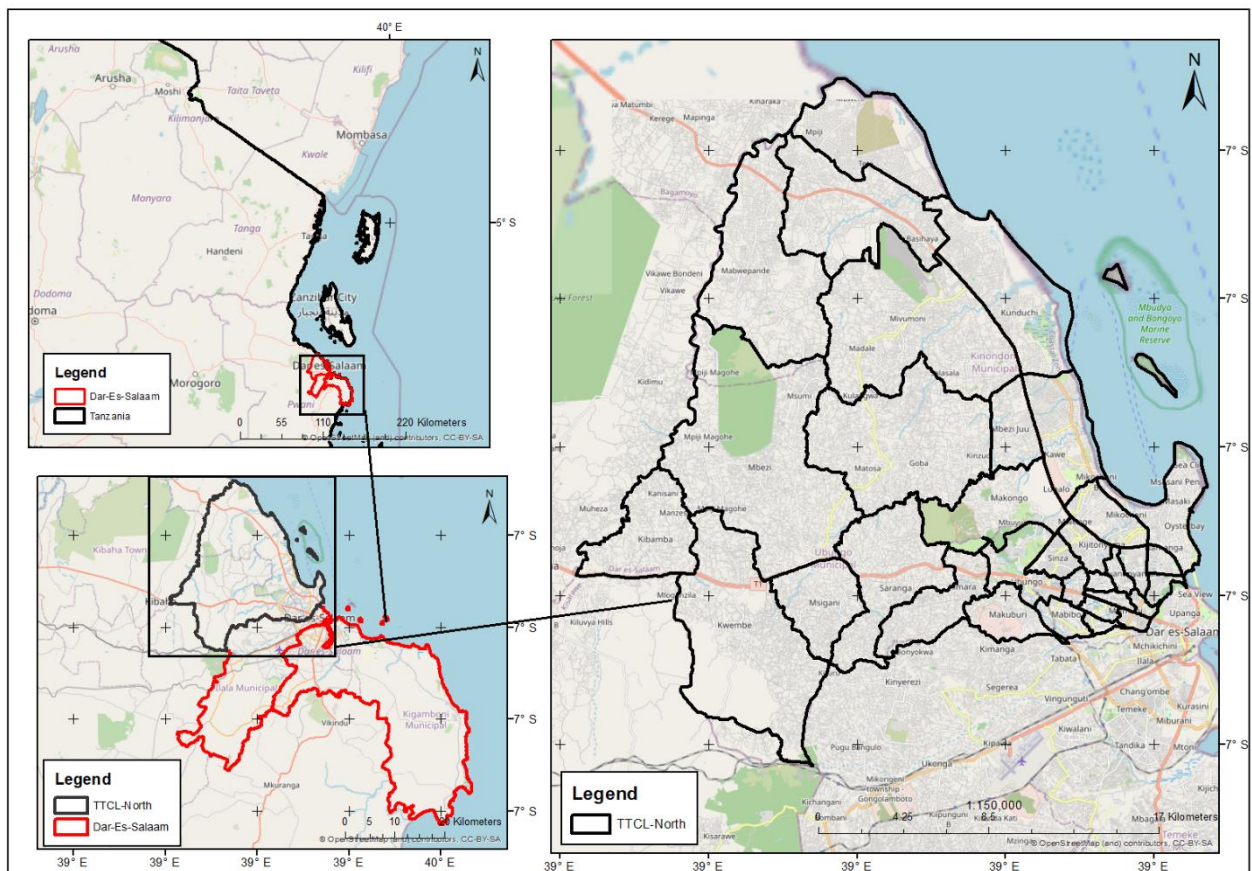


Figure 1: Dar es salaam location map

3.3 Work flow

The workflow for identifying suitable locations for Remote Subscriber Unit (RSU) placement in Dar es Salaam involved a systematic GIS-based approach integrating spatial analysis and multi criteria decision making using model builder. The process began with data collection, the data collected were Digital Elevation Models (DEM), land use/land cover, roads, rivers, and buildings. Next, a restriction model was created using buffering and raster conversion techniques to exclude areas unsuitable for RSU deployment. Then a suitability model was developed by reclassifying key factors (slope, land use, and proximity to roads) and weighted overlay analysis was applied.

The final step involved combining the restriction and suitability models through raster multiplication to identify the most suitable areas for RSU installation. This workflow ensured that site selection is efficient, data-driven, and aligns with technical, environmental, and infrastructural considerations.

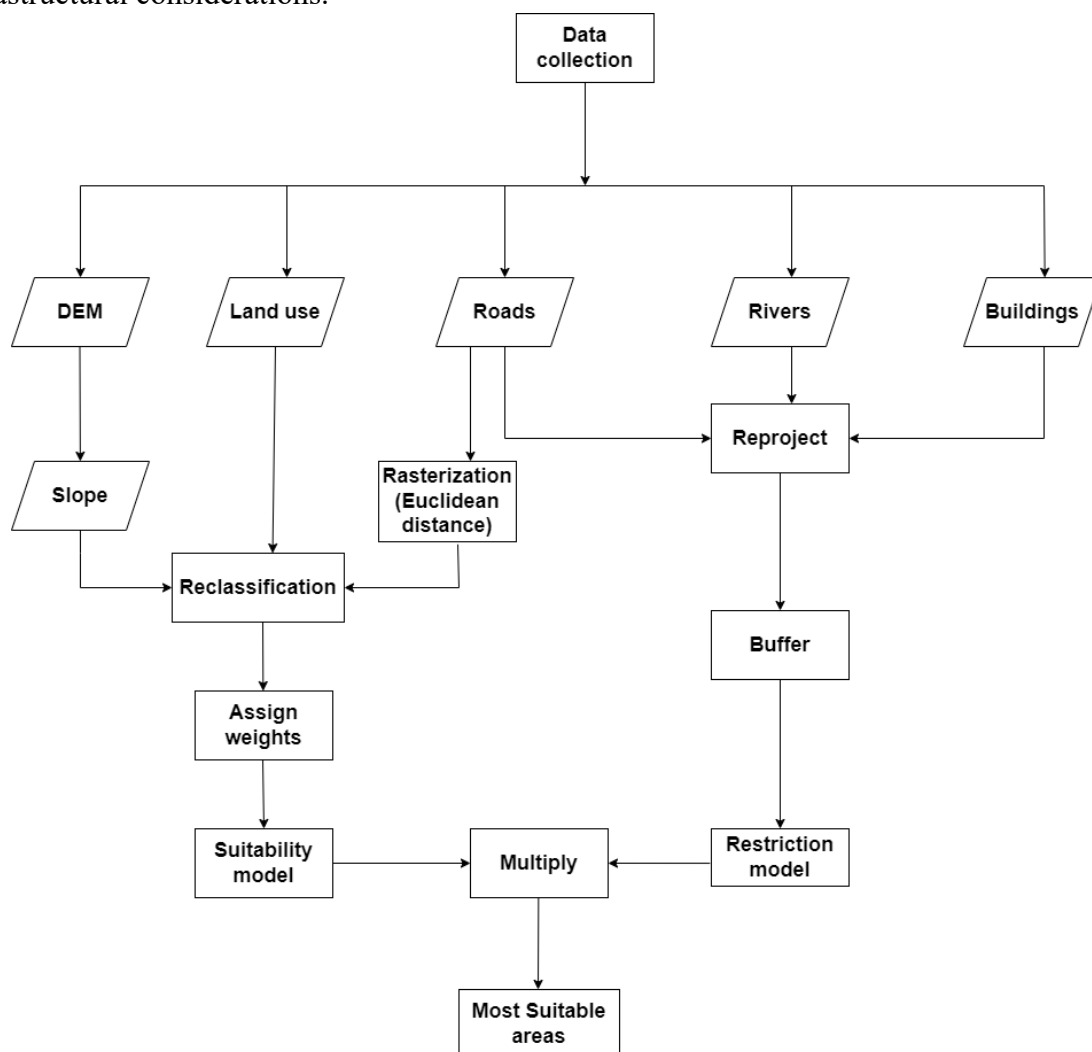


Figure 2:work flow

3.4 Data used

This study utilized a range of spatial datasets to develop a comprehensive GIS-based model for identifying suitable locations for Remote Subscriber Unit (RSU) deployment in Dar es Salaam. Key data included a 30-meter resolution Digital Elevation Model (DEM) for slope analysis, 10-meter resolution Land Use/Land Cover (LULC) data to assess human activity, and road network data from bbbike for evaluating accessibility through Euclidean distance and buffering. River and building data from OpenStreetMap (OSM) were used to define environmental restrictions and proximity to demand centers, respectively.

Table 1: Data used

S/N	DATA	SOURCE	RESOLUTION	FORMAT	USES
1	Digital elevation model	USGS website	30m	Raster	To show elevation of the area
2	Land use and land cover	Esri	10m	Raster	To show activities taking place in the area example agriculture, bare land
3	Roads	BBBike	Exact coordinates	Shapefile (Polygon)	To determine suitable distance RSU should be from roads
4	Rivers	OSM	Exact coordinates	Shapefile (Polygon)	To avoid placing RSU near them
5	Buildings	OSM	Exact coordinates	Shapefile (Polygon)	RSU should near buildings in order to provide suitable service

3.5 Pre-processing

Preprocessing in this study involved several critical steps to prepare the spatial datasets for analysis and ensure compatibility across all layers within the GIS environment. First, each dataset such as the Digital Elevation Model (DEM), land use/land cover (LULC), roads, rivers, and buildings were clipped to match the boundary of the study area, which focused on northern Dar es Salaam. This step ensured that only relevant geographic data was included, reducing processing time and improving analysis focus.

Next, all datasets were projected into a common coordinate system, specifically WGS 84 UTM Zone 37S, to maintain spatial accuracy during overlay operations. For vector layers like roads, rivers, and buildings, buffering was applied: roads were buffered by 200 meters, rivers

by 1000 meters, and buildings by 500 meters. These buffers were designed to define zones of influence or restriction, based on proximity considerations for RSU deployment. The criteria were obtained from TTCL headquarters.

After buffering, vector datasets were converted into raster format to allow for uniform spatial analysis and integration through map algebra tools in GIS. Rasterization was followed by the use of tools like “Is Null”, which helped in creating binary layers (1 for suitable/unrestricted areas and 0 for restricted zones). Additionally, continuous variables like slope and distance to roads were reclassified into standardized suitability classes ranging from 1 (least suitable) to 5 (most suitable), ensuring all inputs could be integrated into a multi-criteria evaluation using weighted overlay techniques.

This preprocessing ensured that each dataset was spatially aligned, logically categorized, and ready for combined suitability and restriction modeling, forming the foundation for accurate RSU site selection analysis.

3.6 Processing

3.6.1 Method used

The method used to deliver the outputs was model builder. Model builder is a visual programming tool that strings(joins) tools and processes into a workflow. It can perform processes like buffering and reclassification. For this dissertation model builder was used to create a restriction model, suitability model and the final output. Below are the symbols used in a model builder.

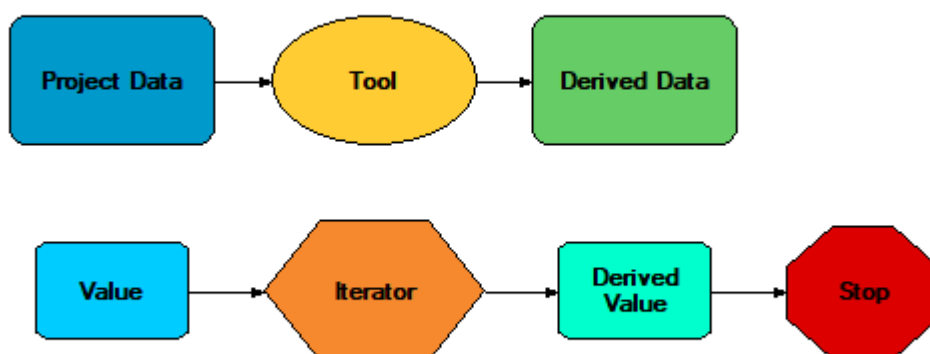


Figure 3:symbols used in a model builder

3.6.2 Creating a restriction model

The restriction model served to eliminate areas that were unsuitable for RSU installation due to environmental risks, inaccessibility, or lack of service demand. This model was built using key spatial layers including rivers, roads, and buildings. Each feature was buffered to define zones of influence: rivers were buffered by 1000 meters to avoid flood-prone areas, roads by 200 meters to ensure access for installation and maintenance, and buildings by 500 meters to ensure proximity to demand centers. After buffering, the vector data were converted into raster format, enabling compatibility for raster-based operations. Using the “Is Null” function in GIS, binary raster layers were created where cells inside the buffer zones were assigned a value of 0 (restricted), and those outside received a value of 1 (unrestricted). These binary rasters were then multiplied together to generate the final restriction model. Only areas that passed all criteria (value = 1) were retained for further suitability analysis, while those failing any condition were excluded.

Final restriction model is given by

$$\text{final restriction} = \text{river restriction} \times \text{building restriction} \times \text{road restriction} \dots (3.1)$$

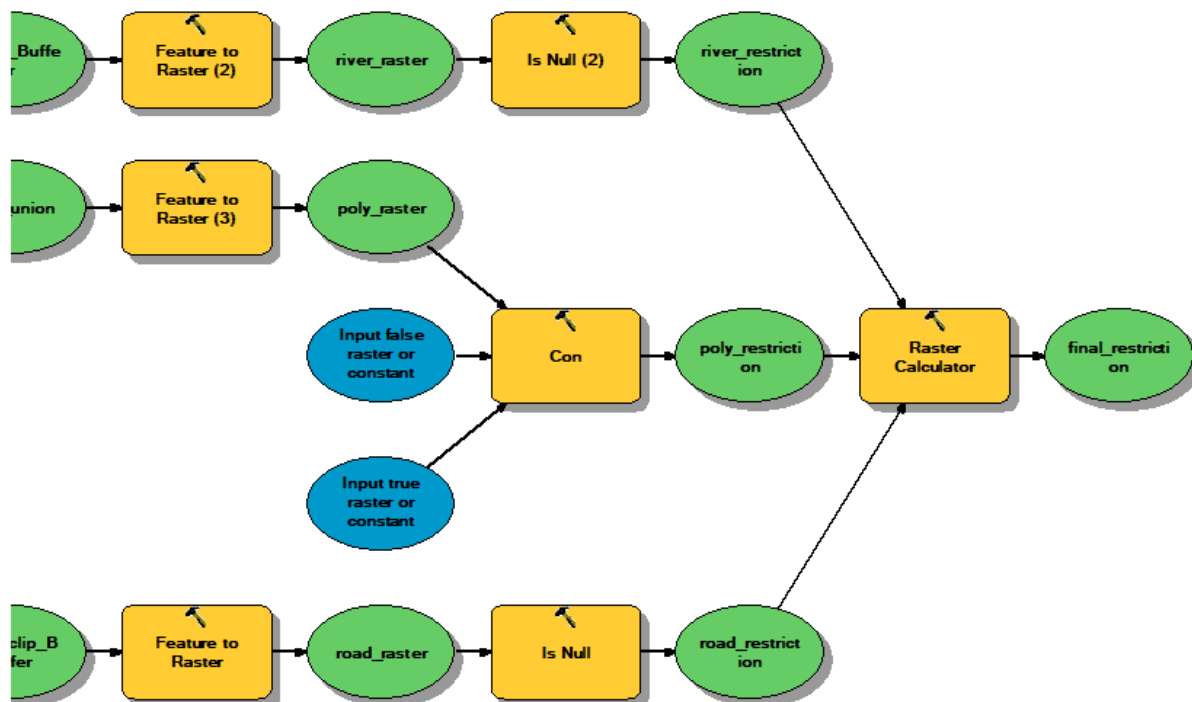


Figure 4: Final restriction model

The full description of figure 4 is found in the appendix page 40.

3.6.3 Creating a suitability model

The suitability model was designed to assess and rank areas based on their appropriateness for RSU deployment, taking into account three key spatial factors: slope, land use/land cover (LULC), and proximity to roads. The process began with reclassification, where each factor was standardized into a common scale of 1 to 5, representing levels of suitability. The slope was derived from the DEM and reclassified so that suitable areas ($0-5^\circ$) received higher scores, while steep areas were not suitable due to construction challenges. Land use data was categorized to favor urban and built-up zones, which indicate higher demand for telecommunication services, whereas water bodies, forests, and agricultural lands were assigned lower suitability scores. For road proximity, Euclidean distance was used to generate a distance raster, which was then reclassified to prioritize areas within 200-500 meters of roads, reflecting the importance of access for infrastructure maintenance.

The reclassified layers were integrated using a weighted overlay, where slope was assigned the highest weight (60%) due to its influence on construction feasibility, followed by land use (25%) and road proximity (15%). The overlay process involved multiplying each layer by its respective weight and summing the results, producing a composite raster that represented overall suitability for RSU placement.

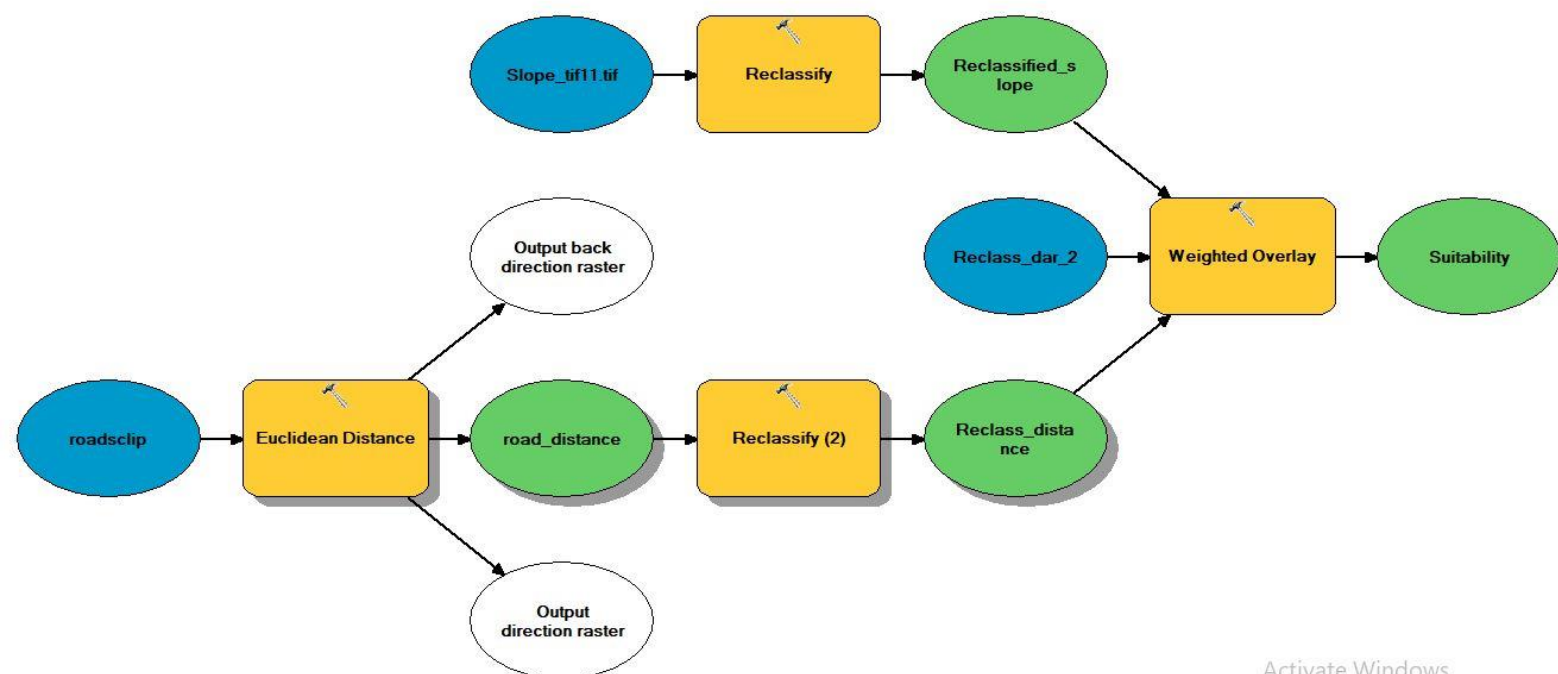


Figure 5: Suitability model

3.6.4 Identifying Suitable areas

After generating both the restriction and suitability models, the final step involved combining them through raster multiplication to isolate areas that were both suitable and unrestricted. The restriction model, a binary raster, acted as a spatial mask by eliminating unsuitable zones (value = 0). When multiplied with the continuous suitability raster (values 1-5), the output preserved only those cells that met all suitability criteria and were not restricted. The resulting final suitability map highlighted specific zones with varying degrees of appropriateness for RSU deployment. This step ensured that site selection recommendations were not only based on environmental and infrastructural favorability but also accounted for real-world constraints such as flood risk and service demand. The output was then used to guide potential RSU locations for field verification and planning purposes.

The two raster's were multiplied using raster calculator:

$$\text{Most suitable areas} = \text{Suitability model} \times \text{Final restriction model} \dots \dots \dots (3.2)$$

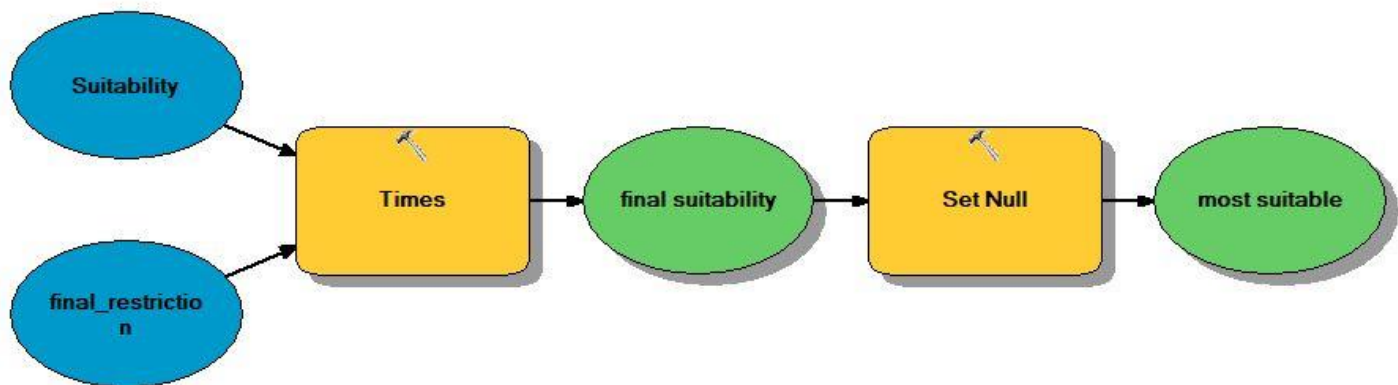


Figure 6: Suitable areas

3.6.5 Validation

To evaluate the accuracy and reliability of the final suitability model, a validation exercise was conducted using existing RSU locations obtained from Tanzania Telecommunications Corporation Limited (TTCL). Eight RSU sites within Dar es Salaam, including those in Ubungo, Sinza, Mikocheni, and Tegeta, were used as reference points. Their geographic coordinates were imported into the GIS environment and overlaid onto the final suitability raster. Using the “Extract Values to Points” tool, the suitability score of each RSU location was retrieved from the raster. These scores were then categorized into three classes: less suitable (1–2), moderately suitable (3–4), and highly suitable (5).

Table 2: Validation table

Extract_csv_Fea3, 6/23/2025, Page 1

type	latitude	longitude	name	Shape *	RASTERVALU	OBJECTID *	Suitability
W	-6.708011	39.206759	africana RSU	Point	3	4	less suitable
W	-6.771326	39.240127	ubungo RSU	Point	4	1	moderately suitable
W	-6.730571	39.212572	tangibovu RSU	Point	4	5	moderately suitable
W	-6.677259	39.204841	kunduchi RSU	Point	4	6	moderately suitable
W	-6.795153	39.219114	Sinza RSU	Point	5	2	highly suitable
W	-6.782357	39.245636	kinondoni RSU	Point	5	3	highly suitable
W	-6.658887	39.183971	Tegeta RSU	Point	5	7	highly suitable
W	-6.753707	39.244249	mikocheni RSU	Point	5	8	highly suitable

$$\text{percentage accuracy} = \left(\frac{\text{number of correct prediction}}{\text{total number of validated points}} \right) \times 100\% \dots \dots \dots (3.3)$$

Number of correct predictions - RSUs falling in moderately or highly suitable areas=7

Total number of validated points - Total RSUs validated=8

$$\text{Percentage accuracy} = \left(\frac{7}{8} \right) \times 100\% = 87.5\%$$

Further analysis, by using the same formula out of 8 validation points:

- ✓ 4 RSUs (50%) fell in highly suitable areas,
- ✓ 3 RSUs (38%) were in moderately suitable areas,
- ✓ 1 RSU (12%) fell in a less suitable area.

Results showed that 87.5% of existing RSUs fell within moderate to high suitability zones, with 50% in highly suitable and 38% in moderately suitable areas. Only one RSU was found in a less suitable zone. This strong alignment between the model’s predictions and real-world deployment validated the robustness of the GIS-based methodology and demonstrated its effectiveness in supporting data-driven telecommunication infrastructure planning.

CHAPTER 4

4 RESULTS AND ANALYSIS

4.0 Overview

This chapter presents the results of the spatial analysis, including land cover maps, slope and elevation maps, proximity buffers, and the final suitability map for RSU placement. It interprets these results by explaining how each criterion contributed to the final output. High-, medium and low-suitability zones are identified and discussed in the context of existing telecommunication infrastructure and population centers in Dar es Salaam.

4.1 Reclassified roads

Euclidean Distance was used to compute the proximity of each pixel to the nearest road. The distance raster was reclassified into five classes. Areas closer to roads (e.g., within 200m) were scored higher (class 5) to ensure easy access for maintenance. Areas far from roads received lower scores, indicating lower priority. This layer contributed to identifying locations that balance technical feasibility and service accessibility.

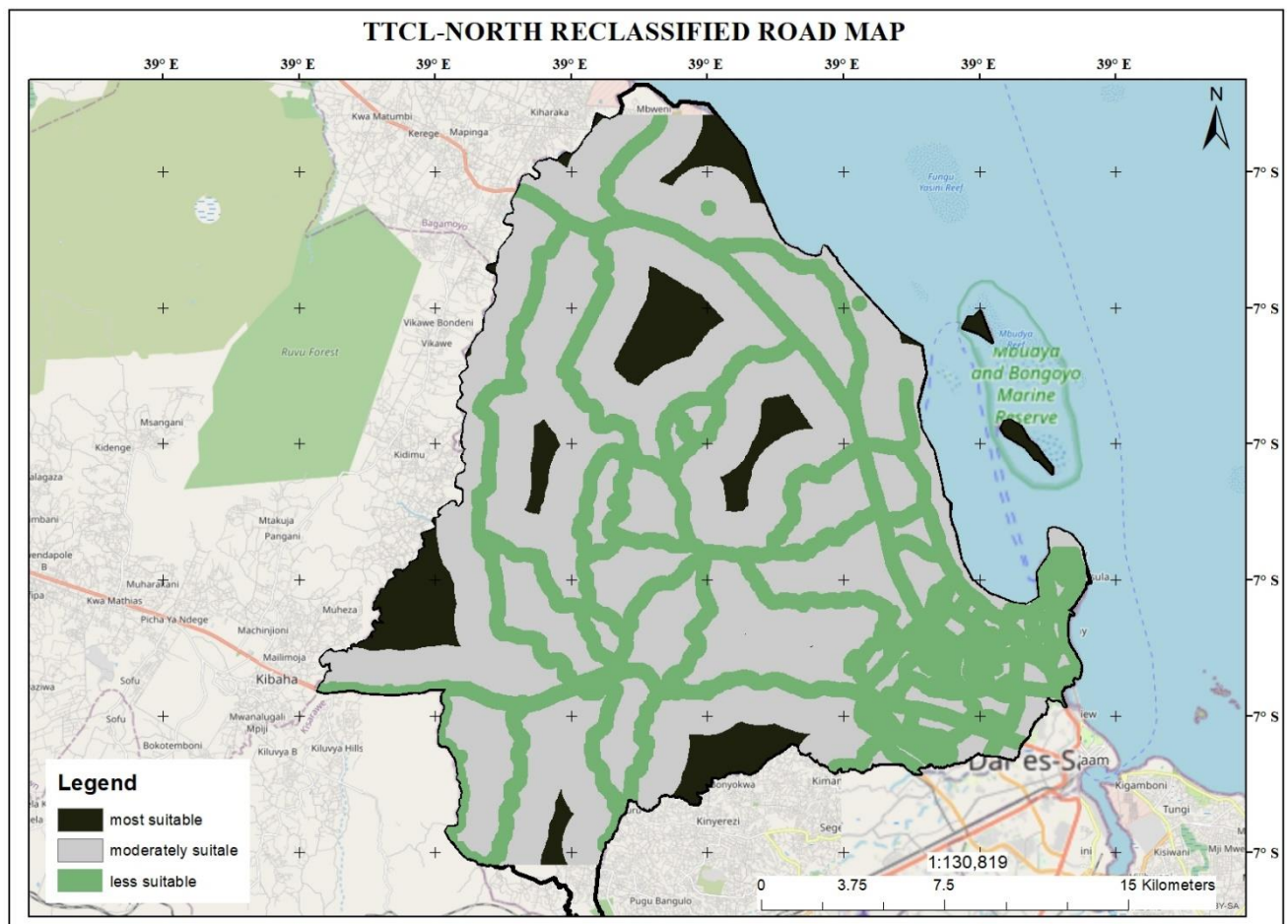


Figure 7: Reclassified roads

This map was generated by combining buffered zones around roads, rivers, and buildings. Using raster conversion and Boolean logic. Areas within restricted buffers were given a value of 0 (unsuitable), Areas outside all buffer zones received a value of 1 (allowed).



The LULC layer was reclassified to assign suitability scores to different land types. Built-up or urban areas were assigned the highest scores (i.e. 5) as they represent zones with high demand for telecommunication services. Water bodies, forests, and agricultural lands received lower scores due to challenges related to infrastructure deployment and low demand. This reclassified map allowed for the integration of land use as a key criterion in the weighted overlay analysis

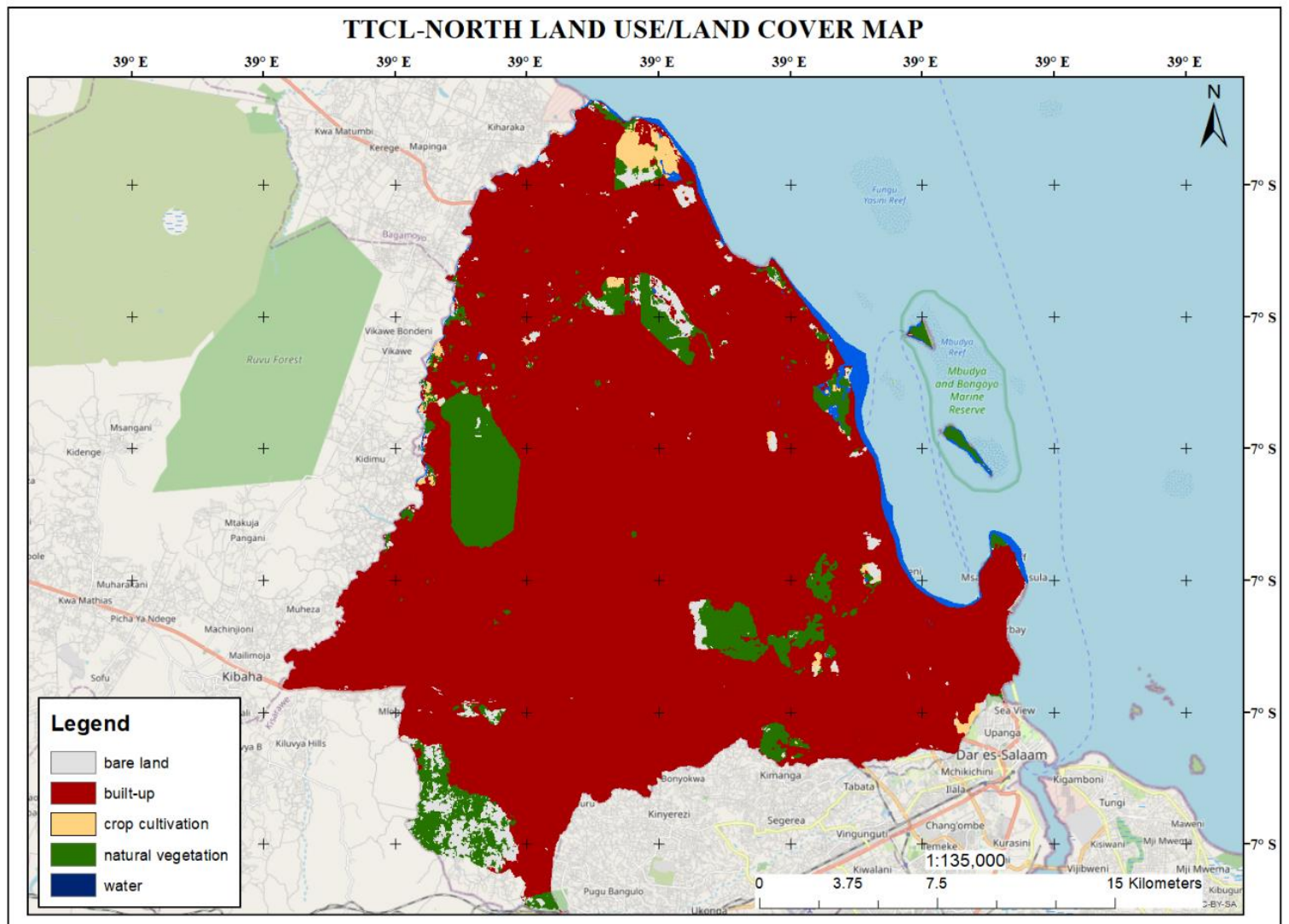


Figure 9:land use/land cover

4.4 Reclassified slope

Slope data was extracted from the DEM (30m resolution) and measured in degrees. The slope was reclassified into five suitability classes. Gentle slopes (0° - 5°) were considered most suitable (class 5) for construction and accessibility. Steep slopes ($>20^{\circ}$) were considered least suitable (class 1) due to difficulty in construction and risk of instability. This output was essential for ensuring RSUs are placed in a favorable terrain.

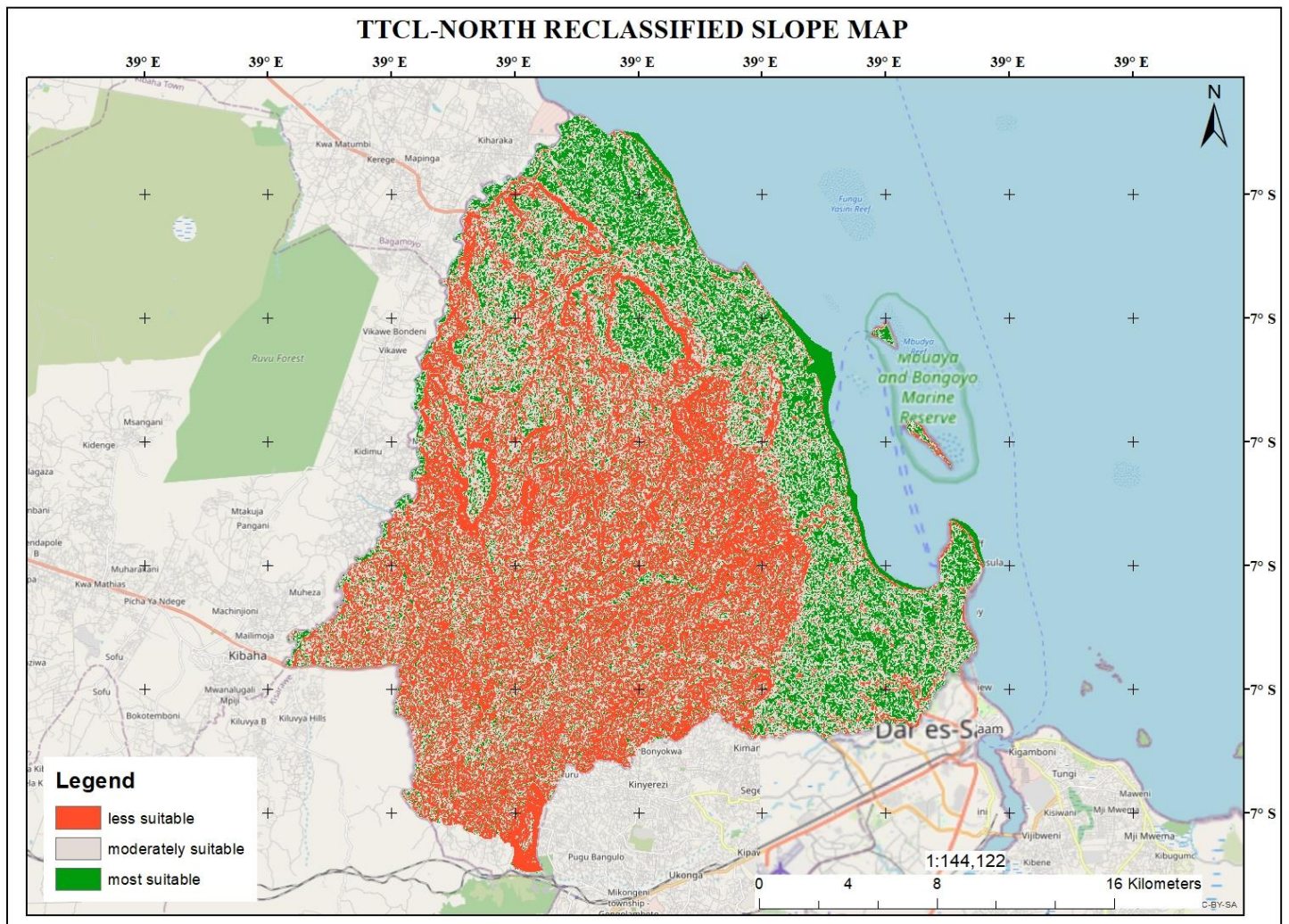


Figure 10: Reclassified slope

4.5 Suitability map

The suitability model was created by integrating the reclassified layers of slope (60% weight), land Use / Land Cover (25%), road Proximity (15%). A weighted overlay analysis was performed to generate a composite raster. The result displays a gradient of suitability (e.g., values from 1 to 5), highlighting areas that best meet the combined criteria. This model identified potentially optimal zones from a technical and demand perspective

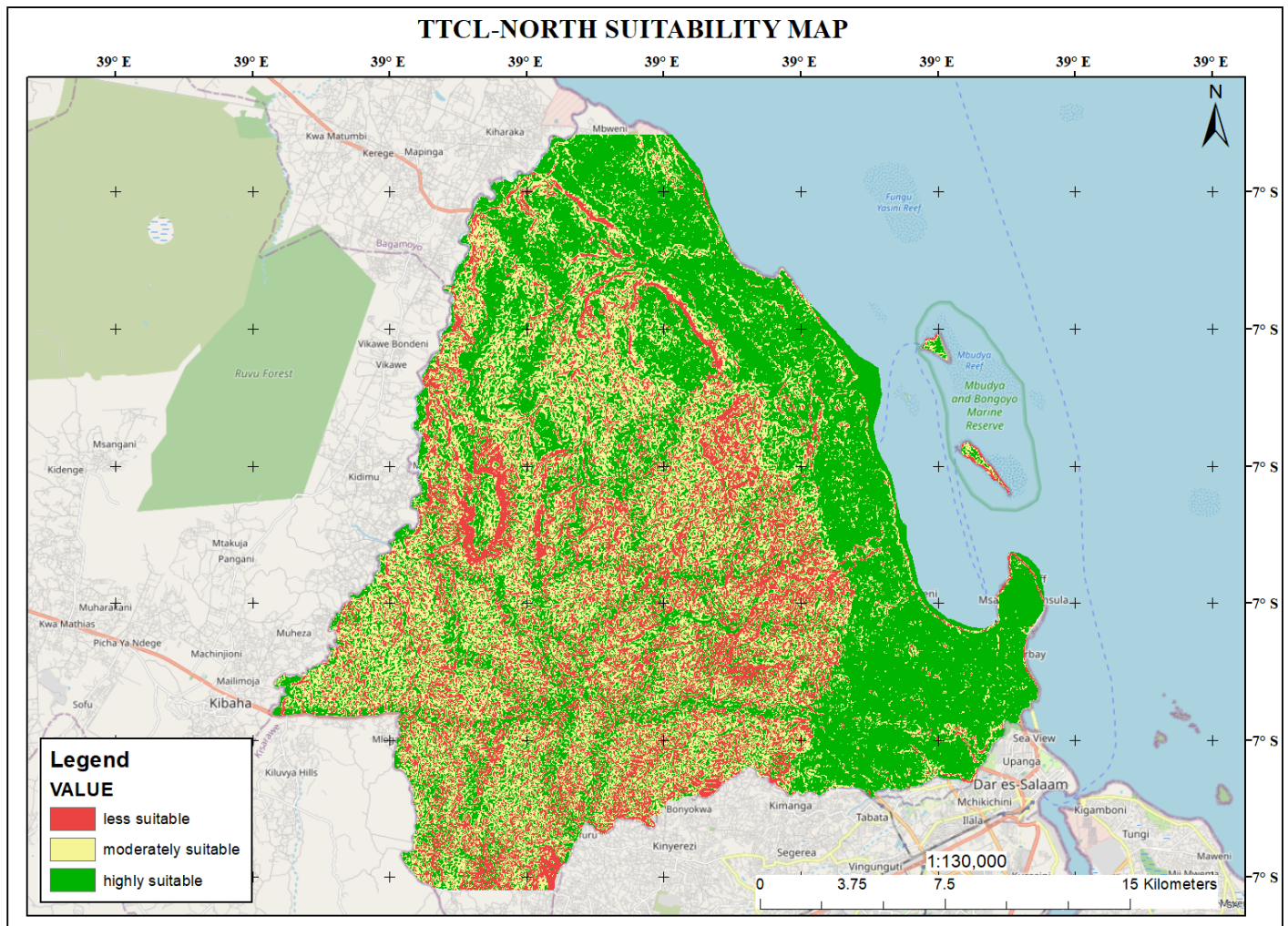


Figure 11: Suitability model

4.6 Final suitability

Final Suitability refers to the output map or raster layer that results from combining the suitability model (which shows how favorable different areas are for RSU placement) with the restriction model (which masks out areas that are not allowed or practical for development). This combination gave a refined result that highlighted only those areas that were both suitable and unrestricted in other words, actually usable for RSU deployment.

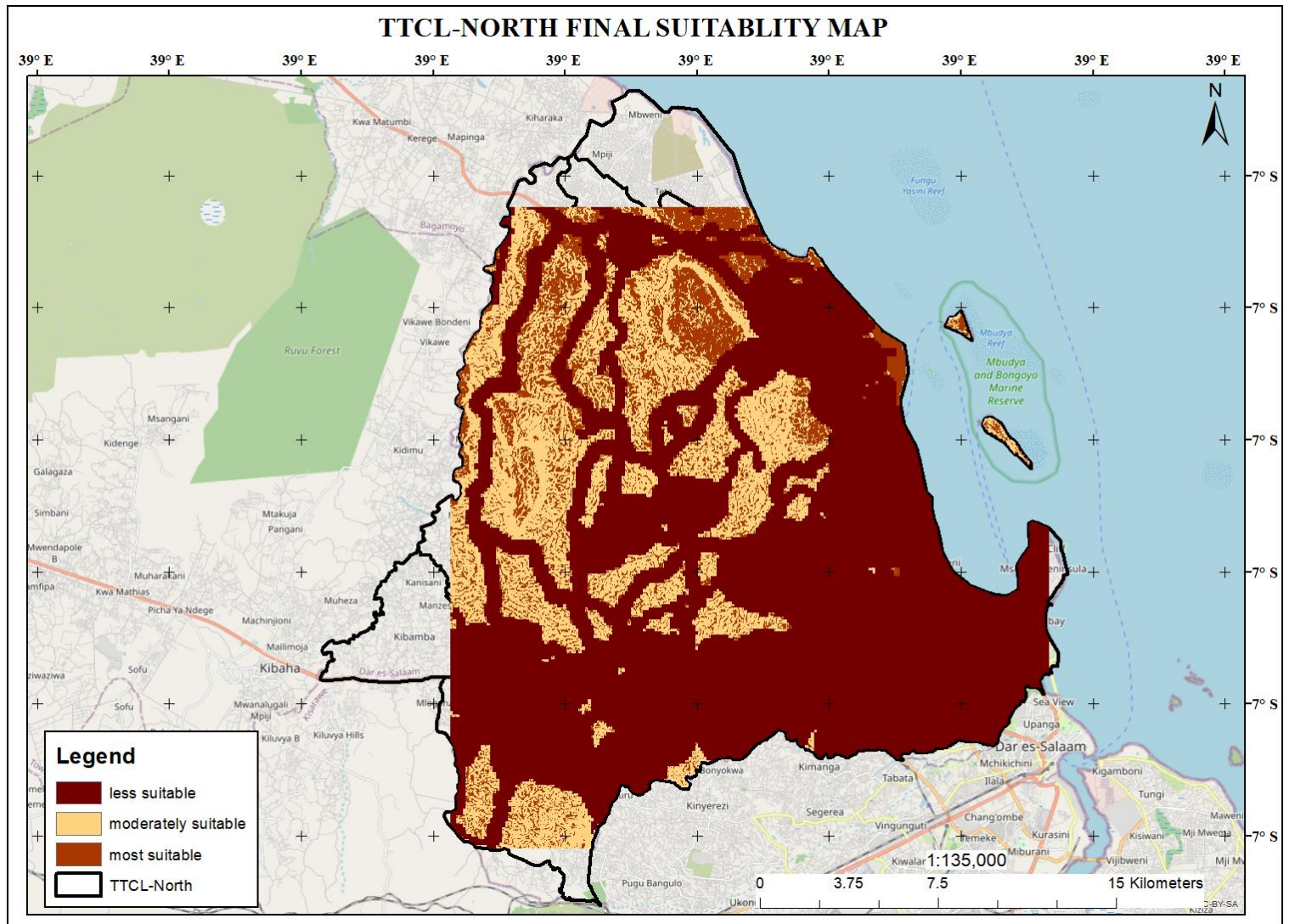


Figure 12: Final suitability map

4.7 Suitable areas

The final suitable areas map was generated by multiplying the suitability model with the restriction model. This ensured that only areas that are both highly suitable and free from restrictions are retained. The resulting map highlights specific geographic zones where RSUs can be strategically deployed. These areas can now be prioritized for field validation, cost estimation, and network planning by TTCL or other service providers

CHAPTER FIVE

5 DISCUSSIONS

This chapter interprets the findings from the results (Chapter 4) and examines their relevance in addressing the research objectives. The discussion also reflects on the strengths and limitations of the methodology, highlights practical implications for telecommunication planning in Dar es Salaam, and assesses how well the model aligns with real-world conditions using validation data.

5.1 Interpretation of Reclassified Thematic Layers

The study utilized three key criteria layers slope, land use/land cover (LULC), and road proximity all reclassified into standardized suitability classes (1 to 5). Each reclassification helped show the influence of specific geographic factors on RSU placement. The approach aligns with the work of (Ustaoglu & Aydinoglu, 2020b) who emphasized the importance of standardizing spatial criteria before multi-criteria analysis for infrastructure planning.

5.1.1 Slope Reclassification

This based on the DEM-derived slope, flatter areas (0° – 5°) were classified as highly suitable. The rationale is that low slopes are easier and cheaper to develop and maintain. In contrast, steep slopes pose construction challenges and potential signal instability. The reclassified slope map revealed that a significant portion of the study area falls within flat to moderately sloped zones, which enhances the potential for RSU deployment. this mirrors the findings of (Murray, 2002), who identified slope as a dominant factor in site selection for communication towers due to its impact on construction cost and signal stability.

5.1.2 Land Use/Land Cover (LULC) Reclassification

Built-up and mixed urban areas received the highest suitability scores due to their high population density and demand for services. Conversely, agricultural, forested, and waterbody areas were given lower scores due to limited access and reduced service demand. This reclassification ensured that the model prioritized areas where telecommunication infrastructure would have the greatest impact. This matches findings by (Baringoltz, 2008), who stressed that telecommunication infrastructure performs best when installed in zones with high human activity and network usage.

5.1.3 Road Proximity (Euclidean Distance) Reclassification

Proximity to roads was analyzed using Euclidean Distance and reclassified to favor locations within 200-500 meters. This was based on the need for maintenance access and reduced deployment costs. This is supported by (Himoto, 2022), who explained that access to infrastructure is essential for long-term maintenance and cost efficiency. The road distance layer complemented the slope and land use criteria by ensuring that highly suitable zones were also physically reachable.

5.2 Analysis of the Restriction Model

The restriction model was created to eliminate zones unsuitable for RSU deployment. Buffering was applied to rivers (1000m), roads (200m), and buildings (500m). The study aligns with (Ustaoglu & Aydinoglu, 2019) who used similar methods to prevent development in high-risk areas. and then converted into raster format and combined using Boolean multiplication. This follows the raster masking strategy suggested by (Zhang, 2010) in spatial decision support systems. The goal was to avoid flood-prone areas near rivers, which pose both environmental and technical risks, locations too far from buildings, which may lack sufficient user demand, zones far from roads, which are hard to access for equipment transport and repairs.

The model effectively masked out these restricted areas. Its integration ensured that only feasible zones were considered in the final suitability analysis. This methodology is consistent with (Moore, 2020), who emphasized the need to avoid ecologically sensitive and flood-prone zones in telecommunications network planning.

5.3 Suitability Model and Weighted Overlay Results

The reclassified layers were integrated using a Weighted Overlay model with the following weights slope-60%, land Use-25%, road distance-15%. These weights reflect the practical importance of each factor, with slope being the most influential due to its effect on infrastructure costs and safety. Similar weighting approaches have been validated in studies by (Rinner & Voss, 2013) and (Ustaoglu & Aydinoglu, 2019b) especially when expert input is not directly available.

The final suitability map showed highly suitable zones clustering in low-slope, built-up, and accessible areas, moderately suitable zones mostly in peripheral residential and semi-developed areas, low-suitability zones occurring in steep, forested, or inaccessible regions.

The output demonstrated that GIS based MCDA could effectively identify target areas for RSU expansion. The weighted overlay method helped balance physical constraints with infrastructure needs, offering a transparent and repeatable planning approach. This mirrors the conclusions of (Ustaoglu et al., 2021) who found that such zones support better signal propagation and user satisfaction. The clustering of high suitability in Dar es Salaam's peri-urban corridors also reflects broader urban expansion trends discussed in East African urban infrastructure planning literature.

5.4 Final suitability map

Final suitability refers to those locations that are not only geographically and environmentally favorable but are also unrestricted by physical constraints (barriers). The suitability map created using reclassified slope, land use, and road proximity was filtered through the restriction model, which masked out high-risk or inaccessible zones. The result is a realistic, deployable map showing where RSUs should be installed to support Dar es Salaam's growing communication needs.

This methodological approach aligns with studies like (Eda Ustaoglu Arif Cagdas Aydinoglu, 2019) and (Sancho-Velasco & Esbert, 2019) who emphasize the need for geospatial modeling in service planning to go beyond theoretical suitability and focus on implementation feasibility especially in urban and semi-urban African settings.

The final map highlights areas such as Sinza, Tegeta, and Ubungo as highly suitable for RSU deployment. These neighborhoods are not only well-connected by roads and infrastructure but also represent high-demand zones with growing populations and increasing digital needs.

This aligns with urbanization trends discussed in (Kombe & Kreibich, 2000) who described how rapid growth in Dar es Salaam is placing pressure on public services, including ICT.

5.5 Identification of Suitable Areas

By multiplying the suitability model with the restriction model, the study eliminated unsuitable zones and retained areas that were both technically appropriate and legally accessible. This method is supported by (Ustaoglu & Aydinoglu, 2019a), who described similar filtering techniques in fiber optic network planning. The final output map highlighted discrete geographic areas, especially in wards such as Sinza, Ubungo, Mikocheni, and Tegeta, as highly suitable for RSU deployment.

These areas are characterized by low slope and flat terrain, high building density (indicating service demand), proximity to road infrastructure. This outcome supports TTCL's infrastructure planning by pinpointing areas with the highest potential for reliable, cost-effective network expansion.

CHAPTER 6

6 CONCLUSION AND RECOMMENDATION

6.0 Overview

This chapter provides a summary of the key findings based on the GIS-based analysis carried out to identify suitable locations for the establishment of new Remote Subscriber Units (RSUs) in Dar es Salaam. The conclusion reflects on how the objectives of the study were met through the methodological steps, and the recommendations offer practical steps that telecommunication planners especially TTCL can adopt. These are directly drawn from the outputs and validation results of the final suitability analysis.

6.1 Conclusion

The study successfully demonstrated that GIS and Multi-Criteria Decision Analysis (MCDA) can be effectively integrated to support RSU site selection in urban environments. Using spatial datasets such as DEM, land use, road networks, rivers, and building footprints, the study developed both a suitability model and a restriction model. The final suitability output, generated by multiplying the two, provided a practical map of areas that are both technically favorable and legally/universally accessible for RSU deployment.

The suitability model identified areas with gentle slope, built-up land use, and proximity to roads as the most favorable. These align with the physical and operational needs of RSU infrastructure. The restriction model effectively excluded flood-prone areas (near rivers), inaccessible zones (too far from roads), and areas with limited-service demand (far from buildings). The final suitability map highlighted specific wards in Dar es Salaam such as Sinza, Ubungu and Tegeta as highly suitable zones for RSU installation. The validation using eight existing RSU coordinates confirmed the model's reliability, with 87.5% of the RSUs located in areas categorized as moderately or highly suitable.

Therefore, the study concludes that GIS-based site suitability modeling provides a reliable, data-driven approach to supporting RSU planning and deployment, reducing the need for costly and time-consuming field surveys.

6.2 Recommendation

Based on the findings and validated results, the following recommendations are provided for practical application and future research:

Adoption of GIS-MCDA in TTCL Planning Workflows. TTCL and other service providers should integrate GIS tools and spatial analysis models into their network expansion strategies. This approach can significantly reduce planning time and improve accuracy in infrastructure rollout.

Prioritize Deployment in Highly Suitable Zones. The areas identified as highly suitable particularly Sinza, Mikocheni, Tegeta, and parts of Ubungu should be prioritized for immediate RSU deployment due to their favorable environmental and infrastructural characteristics.

Engage Experts for Criteria Weighting. The weights used in this study were based on logical assumptions from literature. It is recommended that future models involve telecom engineers, planners, and data scientists to assign more refined weights using methods like AHP (Analytic Hierarchy Process).

Update Spatial Data Regularly. As cities like Dar es Salaam grow rapidly, land use patterns and infrastructure change frequently. Planners should use up-to-date satellite imagery and GIS layers to ensure decisions reflect current on-ground conditions.

Extend the Method to Other Regions. The methodology used in this study can be applied to other urban and rural areas in Tanzania facing similar network coverage gaps. It is scalable and adaptable to different geographies and planning needs.

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APPENDIX

this appendix shows the restriction model formulation using a model builder

